FINAL EXAM INFORMATION

When and where: Monday, December 15, 5:05-7:05 PM, Bardeen 140

Structure of exam: There could be true/false questions and there could be some short answer questions. The rest of the exam will be proof questions similar to those on the homework sets and the activities.

How to prepare: Redoing the activities and homework problems is a good start. It is also a good idea to read the posted solutions for the homework problems. Two old exams are posted to the canvas page for further practice. Note that the structure & point values will be similar to, but potentially not exactly the same as, the old exams. After that, the textbook has additional problems in each section that you could use as practice (selected hints and answers are given in the backs).

Material tested: The final exam will be comprehensive and cover all of the material in the course.

The important Concepts & Learning Objectives for each chapter are given below.

Chapter 0:

- Determine the truth value of a given statement.
- Provide counterexamples for statements that are false.
- Negate specific statements.
- Use a direct proof to prove that a given statement is true.
- Identify logical fallacies or gaps in logic in incorrect proofs.
- Understand and be able to write proofs by cases, contradiction, and contrapositive.
- Identify what proof technique is most appropriate depending on the proposition to be proven.
- State the definition of a set and the empty set.
- Prove one set is a subset of another.
- Prove two sets are equal to each other.
- Understand the set operations of intersection, union, complement, and set minus.

Chapter 1:

- Understand and be able to write proofs by induction. This includes identifying the base case and inductive step required in proofs by induction.

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- Distinguish and understand the relationships between the various sets of numbers, natural numbers, integers, rational numbers, irrational numbers, and the real numbers.
- Prove that a number is irrational using a proof by contradiction.
- Prove properties of the absolute value function using cases.
- State and use the definitions of the supremum and infimum of a set.
- Identify the maximum, minimum, supremum, and infimums of a variety of sets or state they do not exist.
- Know and be able to use the Completeness Axiom, Archimedean Property, and the Density of the rational numbers.

Chapter 2:

- State the formal definition of a sequence.
- State the precise definition of the limit of a sequence and know what it means for the definition not to hold.
- Give a formal proof that an explicit sequence, e.g. $a_n = \frac{1}{n}$ or $b_n = (-1)^n$, converges to a specific number or diverges.
- Give a formal proof that an abstract sequences converges (e.g. prove: if $\lim s_n = 0$ and $s_n \geq 0$ for all n, then $\lim \sqrt{s_n} = 0$).
- Know the formal definition for a sequence to *diverge* to $\pm\infty$ and know how to write a proof showing a given sequence diverges to $\pm\infty$.
- State the definition of a *monotonic sequence*, in particular, what it means for a sequence to be *increasing* or *decreasing* (**Definition 10.1**).
- Determine whether or not a given sequence is monotonic (**Example 10.1**).
- State properties of monotonic sequences related to convergence (**Theorem 10.2**, **Theorem 10.4**).
- Understand what a *subsequence* is (**Definition 11.1**).
- Understand conditions under which subsequences converge or diverge and find those subsequences (**Theorem 11.2**, **Examples 11.3-11.4**).
- State the relationships between sequences and subsequences, including the **Bolzano-Weierstrass Theorem** (**Theorem 11.3-11.5**).
- Understand the definitions of $\limsup$ and $\liminf$ for a sequence (**Definition 10.6)**.
- Understand how the $\limsup$ and $\liminf$ of a sequence relate to the limit (**Theorem 10.7**).
- Understand the definition of a *subsequential limit* and properties of the set of subsequential limits (**Definition 11.6**, **Theorem 11.8**).
- For a given sequence, find its set of subsequential limits and determine its $\limsup$ and $\liminf$ (**Examples 11.5-11.12**).
- State the formal definition of a *Cauchy sequence* (**Definition 10.8**).

- Understand the relationship between convergent sequences and Cauchy sequences (**Theorems 10.9-10.11**).

Chapter 3:

- Know the ϵ/δ definition of $\lim_{x \to a} f(x) = L$.
- Given a concrete function f and concrete numbers a, L, and $\epsilon > 0$, be able to find a $\delta > 0$ that makes the statement "If $0 < |x - a| < \delta$, then $|f(x) - L| < \epsilon$" true.
- Be able to write a ϵ/δ limit proof for linear functions.
- Know the sequence version of $\lim_{x \to a} f(x) = L$.
- Find the *domain* of a given function and a given combination of functions (**Exercise 17.1**).
- Understand the precise definitions of *continuity* for functions, both in terms of sequences (**Definition 17.1**) and in terms of ϵ-δ (**Theorem 17.2)**.
- Use these definitions to determine and prove whether or not a given function is continuous at a specified (or arbitrary) point (**Examples 17.1-17.3**).
- Know what combinations of continuous functions are also continuous (**Theorem 17.3 -17.5**, **Examples 17.4-17.5**).
- Understand the definition of a *bounded function* and be able to determine and prove whether or not a given function is bounded.
- State and know when the *Boundedness Theorem*, *Extreme Value Theorem* and the *Intermediate Value Theorem* apply (**Theorem 18.1-2**).
- Prove that a given function has a *fixed point* or achieves a specific value in a given interval (**Example 18.1**).
- Use the IVT to prove that a polynomial function has a root in a given interval (**Example 18.2**).
- Use the IVT to prove various results involving continuous functions.
- Know the definition of *uniform continuity* and be able to use it to prove whether or not a given function satisfies the property (**Definition 19.1**, **Example 19.2-19.4**).
- Understand the relationship between continuity and uniform continuity (**Theorem 19.2**).

Chapter 5:

- Understand the limit definition of the derivative and how to use it to compute the derivative of concrete functions, like $f(x) = x^2$, $g(x) = \frac{1}{x}$, $h(x) = \sqrt{x}$, etc. (**Definition 28.1**).
- Understand the basic properties of the derivative:
  1. Differentiable $\Longrightarrow$ Continuous (**Theorem 28.1**).
  2. Algebraic rules: sum, product, quotient, chain rule (**Theorems 28.3-28.4**).
  3. The derivative equals 0 at a maximum/minimum point (assuming that the function is differentiable) (**Theorem 29.1**.

- Know the statement of the Mean Value Theorem and the important consequences of the theorem (**Theorems 29.3, 29.5, 29.8**).

Chapter 6:

- Be able to prove a function is integrable or is not integrable using partitions and upper and lower Darboux sums (**Definition 32.1, Examples 32.1-32.2**)
- Understand the relationships between the upper and lower sums of different partitions of an interval (**Theorems 32.2-32.4**).
- Be able to use the Cauchy criterion for integrability to prove that a given function is integrable (**Theorem 32.5**).
- Understand the relationship between integrability and continuity (**Theorem 33.2**).
- Know and be able to use the rules for integration (**Theorems 33.3-33.6**).
- Understand the statements of both parts of the Fundamental Theorem of Calculus (**Theorems 34.1,34.3**)